

碳酸盐岩钙同位素化学分离方法改进及其在青藏高原地质样品中的应用前景

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摘要 为获得更高的测试精度,在前人基础上改进了碳酸盐岩(白云石-方解石)地质样品钙同位素化学分离方案。通过对比云南永善桧溪剖面白云岩-灰岩过渡岩性在不同HCl浓度下淋洗结果,确定了5 mol/L的HCl能有效富集白云石-方解石体系中的钙元素,且热电离质谱(TIMS)测试结果显示4 mol/L的HCl与1.6 mol/L的HCl淋洗方案处理后钙同位素测试结果一致,表明在相同测试精度的情况下,针对白云石-方解石体系下碳酸盐岩地质样品化学分离方案能提高化学分离效率。进一步结合青藏高原研究热点问题,对钙同位素在青藏高原地质样品中的应用前景进行了分析,分析表明,钙同位素在青藏高原隆升前古海洋环境重建、隆升期构造活动恢复及隆升后火成岩来源示踪上具有广泛应用前景。

关键词 碳酸盐岩;钙同位素;化学分离;青藏高原;热电离质谱;古海拔重建;古海洋重建;同位素示踪

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Improvement of Chemical Separation Method of Calcium Isotopes in Carbonate Rocks and Its Application Prospect in Geological Samples of the Qinghai-Tibet Plateau

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Abstract In order to obtain higher test accuracy, the chemical separation scheme of calcium isotopes in carbonate rock (dolomite-calcite) geological samples was improved on the basis of predecessors in this paper. By comparing the leaching results of dolomite limestone transition lithology under different HCl concentrations in Huixi section, Yongshan, Yunnan, it was determined that 5 mol/L HCl effectively

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enriched calcium in dolomite calcite system. In addition, the thermal ionization mass spectrometry (TIMS) test results showed that the calcium isotope test results were consistent after pretreated by 4 mol/L and 1.6 mol/L HCl leaching scheme. This indicated that under the same test accuracy, the chemical separation scheme for carbonate geological samples in dolomite calcite system can improve the chemical separation efficiency. This paper further analyzed the application prospects of calcium isotopes in the geological samples of the Qinghai Tibet Plateau in combination with the hot issues in the study of the Qinghai Tibet Plateau. The analysis showed that calcium isotopes have broad application prospects in the reconstruction of the paleomarine environment before the uplift of the Qinghai Tibet Plateau, the restoration of tectonic activity during the uplift, and the tracing of the source of igneous rocks after the uplift.

Keywords carbonate; calcium isotope; chemical separation; Qinghai-Tibet Plateau; TIMS; ancient elevation reconstruction; ancient ocean reconstruction; isotopic tracing

钙(Ca)元素是构成碳酸盐矿物的主要元素之一^[1-3],广泛分布于地球表层沉积物,在地球物质循环及行星起源研究中起到了重要作用^[4-9]。当前对于钙同位素在高温地质过程中的样品分析测试方法^[10-18]、分馏机理^[15,20-24]以及地质应用分析^[24-28],前人已经进行了大量研究。近年来,随着钙同位素测试技术的发展以及对全球钙循环研究的不断深入^[29-30],钙同位素应用于碳酸盐岩样品代表的古环境、古海洋等领域也得到了众多学者的重视^[31-34]。海洋-大气是碳酸盐岩形成演化的重要环境之一,大气中CO₂收支平衡过程对碳酸盐的溶解—沉淀循环起到了重要控制作用^[29],因此对碳酸盐岩样品中钙元素地球化学行为研究将有助于地质历史时期富钙成岩流体在地球表层循环过程的恢复,对研究碳酸盐岩的示踪具有重要作用。钙同位素在碳酸盐岩领域有着广泛的应用前景。

化学性质来看,Ca属亲石难熔元素,位于第四周期第二主族(IIA),最外层电子构型为4s²,化学性质活泼。从氧化还原性质分析,钙元素仅有氧化态(Ca²⁺),因此不受氧化还原过程影响。钙元素在矿物/矿物间的分配,一定程度上可以视为仅与温度、压力相关,矿物中钙元素可以作为地质温度计或地质压力计^[35]。

Ca同位素体系中包含6种稳定同位素(⁴⁰Ca, 96.941‰; ⁴²Ca, 0.647‰; ⁴³Ca, 0.135‰; ⁴⁴Ca, 2.086‰; ⁴⁶Ca, 0.004‰)和1个放射性同位素(⁴⁸Ca, 0.187‰)。由于⁴⁸Ca半衰期为4.3×10¹⁹a,通常也被视作稳定同位素。由于钙同位素体系包含多种轻同位素,钙同位素分馏δ值具有多种表示方法^[36-37],其中常见的有:

$$\delta^{44/40}\text{Ca}(\text{‰}) = \frac{{}^{44}\text{Ca}/{}^{40}\text{Ca}_{\text{sample}}}{{}^{44}\text{Ca}/{}^{40}\text{Ca}_{\text{standard}}} - 1$$

$$\delta^{44/42}\text{Ca}(\text{‰}) = \frac{{}^{44}\text{Ca}/{}^{42}\text{Ca}_{\text{sample}}}{{}^{44}\text{Ca}/{}^{42}\text{Ca}_{\text{standard}}} - 1$$

其中, $\delta^{44/40}\text{Ca} \approx \delta^{44/42}\text{Ca} \times 2.0483$ 。当前,钙同位素测试中具有多种标准样品,主要包括:NIST SRM 915a 和 IAPSO 标准海水^[38-39]。

钙同位素的测试方法主要有热电离质谱法(TIMs)和多接收电感耦合等离子体质谱法(MC-ICP-MS)。前人研究表明 MC-ICP-MS 在测试钙同位素时存在因以⁴⁰Ar 作为载流气体干扰⁴⁰Ca 测试、样品标样间插法无法校正钙同位素化学分馏、仪器不稳定影响测试精度的缺陷^[10]。TIMs 测试法在双稀释剂技术配合下能获得更高的测试精度。因此,本文立足 TIMs 测试方法和双稀释剂法,在前人^[10]以火成岩地质样品建立的钙同位素化学分离方案的基础上,对比改进碳酸盐岩地质样品钙同位素化学分离方法。

1 实验部分

1.1 主要试剂、仪器及样品

碳酸盐岩地质样品前处理需要使用超纯水、盐酸、硝酸及醋酸,其中,超纯水由 Millipore 公司水纯化系统制备(18.2 MΩ·cm),盐酸、硝酸为优级纯且经 Saville DST 1000 亚沸蒸馏系统二次纯化,醋酸来自 Fisher Scientific 公司,为 LC-MS 级。碳酸盐岩样品前处理所需溶样、接样杯均为 Teflon 材质,经 1:1 优纯级盐酸和 1:1 优纯级硝酸及超纯水密闭 120℃ 蒸煮 24 h,其中 1:1 优纯级硝酸分为二次酸和一次酸,蒸煮顺序为:1)1:1 优纯级盐酸;2)二次 1:1 优纯级硝酸;3)一次 1:1 优纯级硝酸;4)超纯水。实验所需移液管及接液管均经过 5% 稀硝酸和超纯水浸泡。实验离子交换柱采用 Bio-rad Poly-Prep 聚丙烯层析柱(0.8 cm ID×4 cm 高度,10 mL 容量),树脂为 AG MP-50 阳离子交换树脂。

化学测试仪器包括电感耦合等离子体发射光谱(ICP-OES)和 TIMS,ICP-OES 用于淋洗曲线标定

及样品主量元素测试。TIMS 为 Triton plus 型,由 Thermo Fisher Scientific 制造,用于钙同位素测试。

测试样品包括云南永善桧溪剖面寒武系龙王庙灰岩-白云岩样品,NIST SRM 915a 和 IAPSO 标准海水。

1.2 实验设计

利用实际样品与标样建立碳酸盐岩钙同位素前处理及测试方法。为模拟实际样品前处理过程,选取实际样品进行淋滤曲线标定,所选样品为下寒武统龙王庙组白云质灰岩。刘峪菲等^[10]通过对 GBW04412(由文石和方解石组成)进行淋滤曲线标定显示 1.6 mol/L HCl 能有效分离钙元素。针对白云石-方解石体系,在原有 1.6 mol/L HCl 淋滤方案上,重新对多种浓度 HCl 进行淋滤曲线标定。标定完成后进行同位素测试。为避免碳酸盐岩样品中陆源碎屑干扰,实验中所有样品均利用 5% 稀醋酸消解,每次缓慢加入 1 mL,重复 3 次,避免剧烈反应造成样品飞溅。待样品彻底不冒泡后,转移至离心管进行离心,加入双稀释剂后按照改进淋洗方案进行富集,

蒸干后将样品点至 Ta 带放入 TIMS 样品盘,进行同位素测试。具体实验样品及方案设置见表 1、2。实验前,各种试剂和水经空白检测,均符合实验要求。

表 1 云南永善桧溪地区下寒武统
龙王庙组碳酸盐岩样品含量

Table 1 Carbonate sample content list of lower Cambrian
Longwangmiao Formation, Huixi ourcorp, Yongshan
county, Northeastern Yunnan province

样品编号	岩性	CaO/%	MgO/%
hx-3-1B	粉屑泥晶灰岩	50.40	3.32
hx-3-2B	粉屑泥晶灰岩	52.83	2.92
hx-4-1B	含灰白云岩	38.27	17.26
hx-4-2B*	泥晶-粉晶白云岩	32.43	21.34
hx-5-1B	微晶灰岩	51.12	2.54
hx-5-2B	灰质云岩	41.87	9.23
hx-6-1B	泥晶白云岩	31.54	20.03
hx-6-2B	含砂屑泥晶灰岩	49.24	3.93
hx-6-3B	白云质灰岩	45.23	10.55
hx-6-4B	泥-粉晶白云岩	30.62	22.63
hx-6-5B	泥-粉晶灰岩	51.27	1.21

注: * 表示该样品用于淋滤曲线标定。

表 2 云南永善桧溪地区下寒武统龙王庙组碳酸盐岩样品实验步骤

Table 2 Carbonate sample experimental method list of lower Cambrian Longwangmiao Formation,
Huixi ourcorp, Yongshan county, Northeastern Yunnan province

实验编号	样品来源	实验目的
1-1	hx-4-2B	消解后用 4 mol/L HCl 标定淋滤曲线
1-2		消解后用 2.5 mol/L HCl 标定淋滤曲线
1-3		消解后用 2 mol/L HCl 标定淋滤曲线
1-4		消解后用 1.6 mol/L HCl 标定淋滤曲线
2-1	hx-3-1B	TIMS 测试(4 mol/L HCl 淋洗样品)钙同位素值
2-2		TIMS 测试(1.6 mol/L HCl 淋洗样品)钙同位素值
2-3	hx-3-2B	TIMS 测试(4 mol/L HCl 淋洗样品)钙同位素值
2-4		TIMS 测试(1.6 mol/L HCl 淋洗样品)钙同位素值
2-5	hx-4-1B	TIMS 测试(4 mol/L HCl 淋洗样品)钙同位素值
2-6		TIMS 测试(1.6 mol/L HCl 淋洗样品)钙同位素值
2-7	hx-4-2B	TIMS 测试(4 mol/L HCl 淋洗样品)钙同位素值
2-8		TIMS 测试(1.6 mol/L HCl 淋洗样品)钙同位素值
2-9	hx-5-1B	TIMS 测试(4 mol/L HCl 淋洗样品)钙同位素值
2-10		TIMS 测试(1.6 mol/L HCl 淋洗样品)钙同位素值
2-11	hx-5-1B	TIMS 测试(4 mol/L HCl 淋洗样品)钙同位素值
2-12		TIMS 测试(1.6 mol/L HCl 淋洗样品)钙同位素值
2-13	hx-5-2B	TIMS 测试(4 mol/L HCl 淋洗样品)钙同位素值
2-14		TIMS 测试(1.6 mol/L HCl 淋洗样品)钙同位素值
2-15	hx-6-1B	TIMS 测试(4mol/LHCl 淋洗样品)钙同位素值
2-16		TIMS 测试(1.6mol/LHCl 淋洗样品)钙同位素值
2-17	hx-6-2B	TIMS 测试(4 mol/L HCl 淋洗样品)钙同位素值
2-18		TIMS 测试(1.6 mol/L HCl 淋洗样品)钙同位素值
2-19	hx6-3B	TIMS 测试(4 mol/L HCl 淋洗样品)钙同位素值
2-20		TIMS 测试(1.6 mol/L HCl 淋洗样品)钙同位素值
2-21	hx6-4B	TIMS 测试(4 mol/L HCl 淋洗样品)钙同位素值
2-22		TIMS 测试(1.6 mol/L HCl 淋洗样品)钙同位素值
2-23	hx6-4B	TIMS 测试(4 mol/L HCl 淋洗样品)钙同位素值
2-24		TIMS 测试(1.6 mol/L HCl 淋洗样品)钙同位素值
3-1	NIST SRM 915a	监控 TIMS 钙同位素测试质量
3-2	IAPSO	

2 结果与讨论

2.1 淋洗曲线标定

1-1、1-2、1-3、1-4 淋滤曲线的标定结果如图 1 所示。标定结果显示,采用 4 mol/L HCl 进行淋洗时,钙元素淋洗峰在 17 mL 开始析出,27 mL 时钙元素淋洗峰消失。当 HCl 浓度降低至 2.5 mol/L 和 2 mol/L 时, Ti、Sr 元素峰值与钙元素重合。

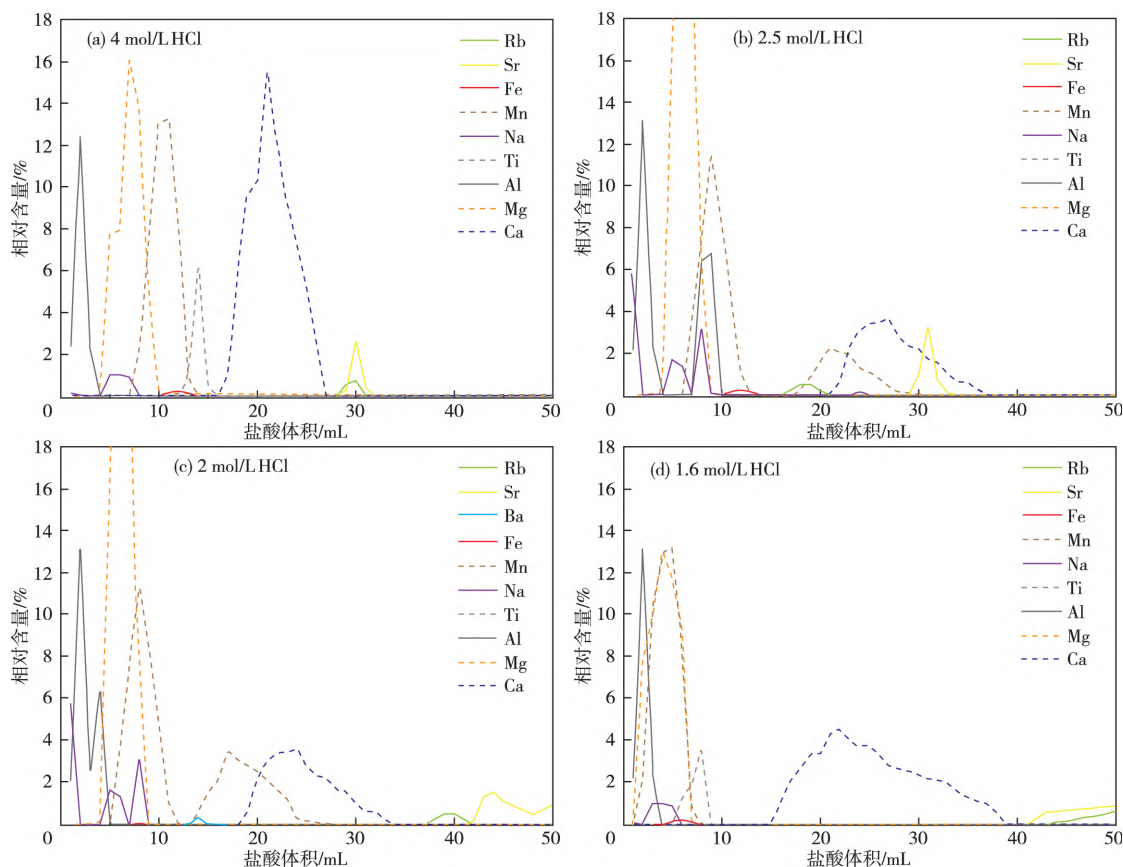


图 1 hx4-2B 样本在不同酸浓度下淋洗曲线

(a) 4 mol/L HCl; (b) 2.5 mol/L HCl; (c) 2 mol/L HCl; (d) 1.6 mol/L HCl

Figure 1 Leaching curves of dolomite-calcite samples under different HCl concentrations

(a) 4 mol/L HCl; (b) 2.5 mol/L HCl; (c) 2 mol/L HCl; (d) 1.6 mol/L HCl

2.2 钙同位素测试结果

测试结果显示,1.6 mol/L HCl 与 4 mol/L HCl 淋洗处理后,钙同位素测试结果接近(图 2),结果表明 4 mol/L HCl 淋洗方案能有效接收钙元素,对测试结果影响较小。4 mol/L HCl 淋洗方案适用于白云石-方解石体系,且钙元素析出峰较窄,相比对 1.6 mol/L HCl 淋洗方案,4 mol/L HCl 淋洗出峰时间更早,出峰浓度区间较小,具有更高的接收

效率。1.6 mol/L HCl 进行淋洗时,钙元素淋洗峰在 15 mL 开始析出,38 mL 时淋洗峰消失,无其他元素重合峰。考虑到 4 mol/L HCl 在实验室使用更为普遍,因此 4 mol/L 浓度 HCl 也可用于碳酸盐岩样品淋洗。为进一步对比 4 mol/L HCl 淋洗方案对钙同位素测试结果影响,对 4 mol/L 及 1.6 mol/L HCl 方案钙同位素测试结果进行了对比。

效率。

对 NIST SRM 915a 和 IAPSO 钙同位素测试结果表明,经化学分离提纯的 NIST SRM 915a 的钙同位素 $\delta^{44/40}\text{Ca} = 0.01 \pm 0.18 (2SD, n = 10)$, IAPSO 的钙同位素 $\delta^{44/40}\text{Ca} = 0.183 \pm 0.14 (2SD, n = 10)$,与文献测试结果一致^[40-41],表明 TIMS 真实反映了两种化学分离方案处理后样品钙同位素结果。

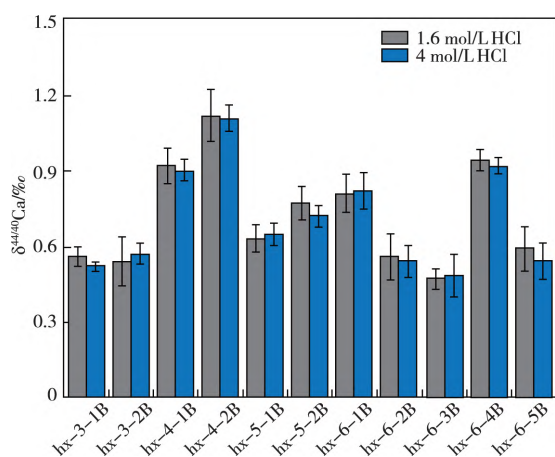


图2 实际样品在不同方案下钙同位素的测定结果对比

Figure 2 Comparison of calcium isotope results from actual samples under different chemical separating methods

3 钙同位素在青藏高原应用前景分析

青藏高原是世界上最高、最新和最广阔的高原,平均海拔在 4 500 m 以上,处于冈瓦纳大陆与欧亚大陆碰撞造山带的关键地带,形成时代较新,高原内部岩浆活动及构造演化极其复杂,其形成演化与特提斯洋俯冲、印度-欧亚板块碰撞密切相关,被国际公认为当今岩石圈研究中的理想地区和大陆动力学研究的“野外实验室”,一直是科学家强烈关注和研究的热点地区^[42-44],钙同位素存在应用前景的重大地质问题包括:印度大陆北漂模型、青藏高原古海拔恢复^[45]。

3.1 印度大陆北漂模型

印度大陆北漂模型是对青藏高原隆升前印度板块与亚欧板块构造运动模式的恢复重建。印度大陆北漂模型包含两种不同模式:1)印度板块与低喜马拉雅、高喜马拉雅和特提斯构造地质单元共同完成北漂(大印度北漂);2)两阶段大陆北漂模型(大印度盆地北漂)。“大印度北漂”模式中,印度板块自南向北依次与低喜马拉雅、高喜马拉雅和特提斯喜马拉雅碰撞形成统一的印度主大陆,统一大陆伴随新特提斯洋盆的俯冲-闭合过程逐渐向北俯冲,与欧亚板块产生碰撞^[46-48]。“大印度盆地北漂”模式中,印度板块主体与低喜马拉雅地块形成的大陆并未继续与高喜马拉雅地块产生碰撞,低喜马拉雅地块与喜马拉雅微陆块(高喜马拉雅与特提斯喜马拉雅的组合体)间发育大印度盆地^[49]。50 Ma 期间,伴喜马拉雅微陆块首先与欧亚大陆碰撞^[49]。25~20 Ma 期间,印度板块主体大陆随着大印度盆地的关闭与欧亚板块产生碰撞^[49]。

许志琴等^[45]将印度大陆北漂模型争议总结为对大印度板块大小划分的差异,认为新特提斯洋盆消减过程中从洋内俯冲到洋-陆碰撞转换的存在性是解决印度大陆北漂模型争议的关键之一。钙同位素作为深部碳循环的有效示踪手段,因地表沉积物钙同位素与地幔钙同位素的显著差异,能有效对洋内俯冲程度进行刻画^[28]。洋-陆碰撞过程中以俯冲、低温高压变质及仰冲构造作用为特征,发育低温榴辉岩套^[50]。钙同位素作为温压条件的重要反映指标,能有效实现地质样品的温压重建,因此在同时对同一地质样品进行洋内俯冲物质循环模拟和温压模拟后,通过比较物质循环中温压条件与低温环境的差异性(低温榴辉岩套),即可对地质样品经过的洋内俯冲、洋-陆碰撞过程进行判断。因此,钙同位素在解决印度大陆北漂模型争议时因其在俯冲和碰撞过程中同位素行为的差异性具有一定的应用前景。

3.2 青藏高原隆升的时限和差异性

青藏高原隆升时限和差异性问题旨在恢复青藏高原的古海拔、重建青藏高原隆升历史,进而揭示青藏高原造山的动力学机制以及高原隆升过程中气候-环境耦合效应。大量研究结果表明,组成青藏高原的各个地块在隆升过程中存在明显的差异性,但对于差异性的具体表现仍存在争议。当前,存在多种恢复青藏高原古海拔的研究手段,包括:钾质火山岩喷发重建、古生物共存法、叶相法、叶片气孔密度法、玄武岩气孔法、氧同位素重建法。

钾质火山岩是钾元素质量百分数大于钠元素质量百分数的火山岩^[51],岩浆来源为岩石圈地幔,其喷发代表了高原隆升至现今海拔^[52]。通过对钾质火山岩进行年代学分析,青藏高原在 13 Ma 时就已经达到了现今海拔高度^[53]。但研究同时表明,距今最古老的钾质火山岩能进一步追溯到 55~45 Ma,表明青藏高原在约 50 Ma 即已经隆升至现今海拔高度,与前者存在明显矛盾,因此,钾质火山岩对古海拔演化推断的可信度存疑^[51,53]。

古生物共存法主要研究对象为植物化石的残体,通过对与化石最为接近的现代植物种属的温度变化范围定性描述古温度特征,从而实现对古海拔的定性恢复^[54]。然而,由于植物化石的鉴定误差或演化过程中植物对栖息地环境适应性改变,利用古生物共存法进行古海拔恢复的可靠性受到了学者质疑^[55-57]。

叶相法与叶片气孔密度法原理相似,均为利用相似气候条件下植物形态学特征较为接近的原理对

古海拔进行重建。目前,基于 Climate-leaf analysis multivariate program (CLAMP) 实现了误差小于 1°C 的年平均气温重建,根据年平均气温重建的古海拔具有较高的精度^[55]。叶片气孔密度法利用了植物化石中气泡密度这一叶相特征对二氧化碳分压进行重建,进而间接还原古海拔,但由于叶子发育的背光、向光性差异及海洋储库中二氧化碳分压的不确定性,利用叶片气孔密度法古海拔进行重建的过程中存在多种不确定因素^[58]。

玄武岩气孔法的基本原理为玄武岩熔岩流顶部气孔压力仅来自于大气压,底部压力来自于大气压与熔岩流重力,因此利用大气压与高程的相关关系可重建古海拔^[59-60]。但由于气泡来源差异及熔岩流冷却后厚度的变化,造成玄武岩样品采集难度较高,玄武岩气孔法运用范围有限^[61]。

利用氧同位素对古海拔的重建原理为大气降水过程中氧同位素随着山脉海拔增加而富集轻同位素^[62],因此利用氧同位素分馏差异即可对古海拔进行重建。但受限于成岩作用改造等因素,氧同位素值会有较大变化,影响古海拔重建^[63]。

由于氧同位素的广泛适用性,SHEN 等^[64]提出基于瑞利分馏与海拔关系的古海拔重建方法,这一方法的条件之一是碳酸盐岩形成的初始温度。利用氧同位素仅能实现对温度的估计,钙元素不受氧化还原过程影响,在一定程度上以温度、压力的相关函数,利用钙同位素能有效计算碳酸盐岩形成的初始温度,结合氧同位素测试结果,能有效提高古海拔重建精度。由于钙同位素对温压条件的响应,在明确各储库钙同位素特征以及钙同位素随海拔变化产生的分馏关系后,可以直接利用钙同位素重建古海拔。因此,钙同位素在青藏高原古海拔重建研究中具有重要的研究价值。

3.3 火成岩岩浆源区示踪

青藏高原形成的隆升作用研究中,印度-欧亚板块碰撞后的陆内汇聚阶段的研究是及其重要的,这一阶段下青藏高原内部的构造演化及其复杂且发育多起岩浆活动,内部各地块之间发育三条大型缝合带,其发育蛇绿混杂岩带及高压-超高压变质代表不同时期的特提斯洋俯冲消亡洋壳残片^[65-67]。尽管目前对于青藏高原地区钾质-超钾质岩石的研究取得了一些成果^[68-70],但是对于特提斯构造域的其他地区而言,青藏高原的复杂性和特殊性造成对富钾岩石的认识依然不足,包括对于超钾质岩浆演化过程于地幔源区估计特征的争议^[71-74];对于与幔源超

钾质岩石同时期的其他后碰撞岩石在成因解读上的争议^[75-78];关于后碰撞岩浆活动所反映的藏南中新世深部动力学过程的争议^[79-81]。而钙同位素作为非传统同位素示踪手段逐渐应用到地球科学中,在示踪青藏高原超钾质火山岩的源区形成于大洋俯冲过程相关具有极大优势,如刘峪菲等^[44]对青藏高原南部拉萨地块中西部米巴勒和麦嘎地区超钾质岩 Ca、Mg 同位素进行了研究,确定了其源区可能有来自俯冲新特提斯洋壳的碳酸盐沉积物加入。也可以作为研究钾质岩、超钾质岩及埃达克岩成岩关系的地球化学手段,如刘峪菲等^[44]应用 Ca、Mg 同位素的研究发现对青藏高原拉萨地块中-南部钾质岩及埃达克岩这两类岩石具有不同的源区组成。因此钙同位素方法能对青藏高原的火成岩岩浆源区进行示踪。

3.4 古海洋重建

除火成岩及变质岩外,青藏高原各个地块中均发育有碳酸盐岩。通过钙同位素可以对碳酸盐岩成岩作用进行评估,有助于解读其所记录的古海洋环境信息,对了解青藏高原的隆升作用具有重要意义。其次由于青藏高原处于一个广泛被海洋淹没时期,特别是上三叠系从青海、藏北、藏南、川西均属于同一生物及沉积环境地理大区^[62],生物群组面貌几乎相当,并且在各时期地层中广泛发育,因此青藏高原中丰富的钙质生物可以广泛应用通过钙同位素重建深时海水钙同位素值。如利用腕足类生物化石重建了奥陶纪至白垩纪海水的钙同位素演化特征^[47],并认为其主要受控于“文石海-方解石海”的演变^[67]。因此,青藏高原古海洋环境研究、青藏高原深时海水钙同位素地质的重建等具有重要意义。

4 结论

本文基于前人建立的碳酸盐岩地质样品钙同位素前处理方法,结合实际碳酸盐岩(白云石-方解石体系)样品,改进了钙同位素化学前处理流程,改进后的测试方案在 TIMS 上的测试结果与改进前的方案基本一致,表明新建立的碳酸盐岩钙同位素测试方案能有效测得样品的钙同位素结果。改进后的化学处理流程相较原流程出峰更早,出峰范围更窄,且 4 mol/L HCl 更容易配制,因此,改进的前处理流程更适用于白云石-方解石碳酸盐岩体系。并结合青藏高原研究的热点问题对钙同位素在青藏高原地质样品中的应用前景进行分析。针对火成岩,钙同位素能对青藏高原陆壳俯冲、火成岩源区示踪进行有效分析。针对碳酸盐岩,钙同位素能对青藏高原隆

升前-隆升过程中古海洋、古环境信息重新解构,配合隆升期构造活动、火山活动的地质样品的钙同位素分析结果,对青藏高原隆升前后一系列地质事件进行整体重建。

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